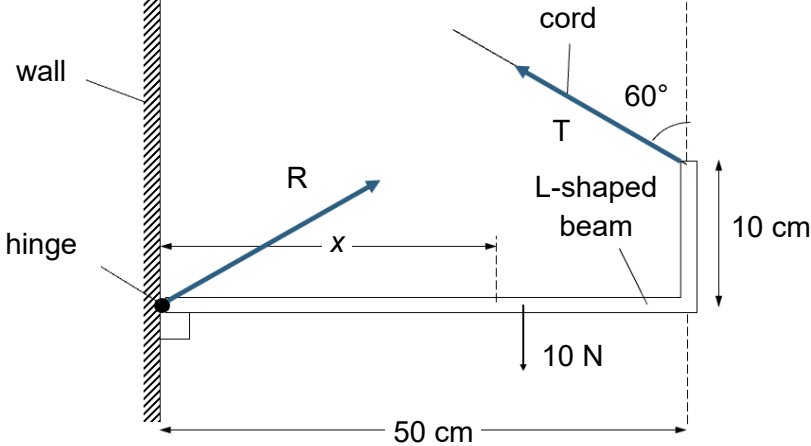


2	(a)	Using hinge as pivot: Clockwise moment due to weight = anti-clockwise moment due to tension Moment due to $T_y = T (\cos 60) 50$ Moment due to $T_x = T (\sin 60) 10$ $Wx = T (\cos 60) 50 + T (\sin 60) 10$ $10x = 5.0 [50(\cos 60) + 10(\sin 60)]$ $x = 16.8 = 17 \text{ cm}$ Note: if either moment vertical/horizontal is not considered, max 1 mark.	M1 A1
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	(b)	(i)	 Both vectors drawn and labelled 3 forces have common point	B1 B1
	(b)	(ii)	$R_x = T_x = 5.0 \sin 60 = 4.33 \text{ N}$ $R_y + T_y = W$ $R_y = W - T_y = 10 - 5.0 \cos 60 = 7.5 \text{ N}$ $ R = \sqrt{R_x^2 + R_y^2} = \sqrt{4.33^2 + (-7.5)^2}$ $= 8.7 \text{ N}$	M1 M1 A0